

Lecture 2 Task: Binary-Coded Genetic Algorithm

Task title: One Generation of Binary-Coded GA for Function Minimization

Consider the function $f(x) = x^2$ where $-5 \leq x \leq 5$. Use a binary-coded genetic algorithm with: chromosome length = 5 bits, population size = 6, crossover probability $p_c = 0.8$, and mutation probability $p_m = 0.1$.

Initial population

Solution	Chromosome
1	00101
2	01011
3	10100
4	11110
5	01101
6	10010

Decoding rule

First decode each binary string into decimal value $DV(s)$, then compute: $x = -5 + (10/31) DV(s)$.

Your task

1. Decode all 6 chromosomes and find the corresponding value of x .
2. Compute the objective value $f(x) = x^2$ for each solution and identify the best solution in the initial population.
3. Use binary tournament selection to create a mating pool of 6 solutions.
4. Pair the selected solutions and apply single-point crossover using $p_c = 0.8$.
5. Apply bit-flip mutation using $p_m = 0.1$.
6. Decode the offspring, compute their objective values, and using survivor selection choose the best 6 solutions for the next generation.
7. Write one short paragraph answering: Did crossover improve all offspring? Did mutation help or hurt? Why is survivor selection important?
8. Run the Python code provided in class for this same problem and include: the output of the initial population, the best solution found, the objective value of the best solution, and the convergence plot or a screenshot of the result.

Submission instructions

- Do all calculations by hand using pen and paper.
- Scan your handwritten solution clearly.
- Attach the Python results as part of the same submission.
- Submit one single PDF file containing: scanned handwritten calculations, short paragraph answer, and Python output/results.

Note: This is a minimization problem, so the solution with the smallest objective value is the best.